

§ 3.2

P1. Solution:

$$a) A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix} \xrightarrow{t_2=1} \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix}$$

$$\xrightarrow{t_3=1} \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = U$$

Pivot variables: x_1, x_3 free variables: x_2, x_4, x_5

$$b) B = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 8 & 8 \end{bmatrix} \xrightarrow{t_2=2} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \end{bmatrix} = U$$

Pivot variables: x_1, x_2 free variables: x_3

P2. Solution:

(a) (i) Set $x_2=1, x_4=x_5=0$, Solve $UX=0$

$$x_1 + 2x_2 + 2x_3 + 4x_4 + 6x_5 = 0$$

$$x_3 + 2x_4 + 3x_5 = 0$$

$$\Rightarrow \begin{cases} x_1 + 2x_3 = -2 \\ x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -2 \\ x_3 = 0 \end{cases}$$

$$\Rightarrow S_1 = (-2, 1, 0, 0, 0)$$

(ii) Set $x_2=0, x_4=1, x_5=0$, then

$$x_1 + 2x_3 = -4 \Rightarrow \begin{cases} x_1 = 0 \\ x_3 = -2 \end{cases}$$

$$x_3 = -2 \Rightarrow x_3 = -2$$

$$\Rightarrow S_2 = (0, 0, -2, 1, 0)$$

(iii) Set $x_2=x_4=0, x_5=1$, then

$$x_1 + 2x_3 = -6 \Rightarrow \begin{cases} x_1 = 0 \\ x_3 = -3 \end{cases}$$

$$x_3 = -3 \Rightarrow x_3 = -3$$

$$\Rightarrow S_3 = (0, 0, -3, 0, 1)$$

(b) set $x_3=1$, solve $UX=0$

$$2x_1 + 4x_2 + 2x_3 = 0$$

$$4x_2 + 4x_3 = 0$$

$$\Rightarrow \begin{cases} 4x_2 + 2x_3 = -2 \\ 4x_2 + 4x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_2 = -1 \\ x_3 = 1 \end{cases}$$

$$\Rightarrow S_4 = (1, -1, 1)$$

P4. solution:

$$a) U = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = R$$

$$b) U = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & -2 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{/4} \begin{bmatrix} 2 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$$

$$N(R) = N(U) \text{ true!}$$

P9. solution:

a) False

b) Yes

c) Yes

d) Yes

P10. solution:

a) impossible

$$b) A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 2 & 4 \end{bmatrix}$$

$$c) A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$$

$$d) A = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

P16. solution: 5 pivots

P26. solution:

$$A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

$$C(A) = N(A) = \left\{ \begin{bmatrix} a \\ -a \end{bmatrix} \right\}$$

P32. solution

$$c) N = \begin{bmatrix} 2 & 3 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow R = [1-2-3]$$

$$(i) N = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \rightarrow R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$(ii) N = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \rightarrow R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Ex 3.3.

P1. solution:

(a) Yes

(b) No

(c) Yes

(d) No

P3. Solution:

$$(i) A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 2 & 4 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 4 & 6 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(ii) B = [A \ A] \rightarrow [R_A \ R_A] \quad \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R_A$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} = R_B$$

$$(ii) C = \begin{bmatrix} A & A \\ A & 0 \end{bmatrix} \rightarrow \begin{bmatrix} A & A \\ 0 & -A \end{bmatrix} \rightarrow \begin{bmatrix} R_A & R_A \\ 0 & -R_A \end{bmatrix} \rightarrow \begin{bmatrix} R_A & R_A \\ 0 & R_A \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} R_A & 0 \\ 0 & R_A \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} = R_C$$

P7. solution:

$$(i) R = \begin{bmatrix} 0 & 0 & 2 & 3 \\ 0 & 0 & 4 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow N = \begin{bmatrix} -2 & -3 \\ -4 & -5 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\Rightarrow Rx=0$ has two special solution.

$$S_1 = \begin{bmatrix} -2 \\ -4 \\ 1 \\ 0 \end{bmatrix} \text{ and } S_2 = \begin{bmatrix} -3 \\ -5 \\ 0 \\ 1 \end{bmatrix}$$

$$(ii) R = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow N = \begin{bmatrix} 1 & 0 \\ 0 & -2 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow S_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad S_2 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

$$P10. (i) A = \begin{bmatrix} 3 & 6 & 6 \\ 1 & 4 & 8 \end{bmatrix}$$

$$U = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} \quad V = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \rightarrow A = UV^T$$

$$(ii) A = \begin{bmatrix} 2 & 2 & 6 & 4 \\ -1 & -1 & -3 & -2 \end{bmatrix}$$

$$U = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad V = \begin{bmatrix} 1 \\ 1 \\ 3 \\ 2 \end{bmatrix} \rightarrow A = UV^T$$

P15. solution:

$$i) \text{rank}(AB) = 1 \text{ where } AB = \begin{bmatrix} 8 & 4 & 16 \\ 16 & 8 & 32 \end{bmatrix}$$

$$ii) AM = \begin{bmatrix} 1+2c & b+2bc \\ 2+4c & 2b+4bc \\ 1+2c & b(1+2c) \\ 2c(1+2c) & 2b(1+2c) \end{bmatrix}$$

$$a) \text{ when } C \neq -\frac{1}{2}, \text{rank}(AM) = 1$$

$$b) \text{ when } C = -\frac{1}{2}, \text{rank}(AM) = 0$$

P20. solution:

A an 2×3 matrix

B an 3×2 matrix

$\Rightarrow AB$ is an 2×2 matrix and BA is a 3×3 matrix

clearly, $\text{rank}(A) \leq 2$ and $\text{rank}(B) \leq 2$

Suppose $BA = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then $\text{rank}(BA) = 3$

but we also have

$$\text{rank}(BA) \leq \text{rank}(B) = 2 \rightarrow \text{contradiction.}$$

$\Rightarrow BA \neq I$

An example: $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$

$$\text{then } AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, BA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \neq I_3 = I_2$$